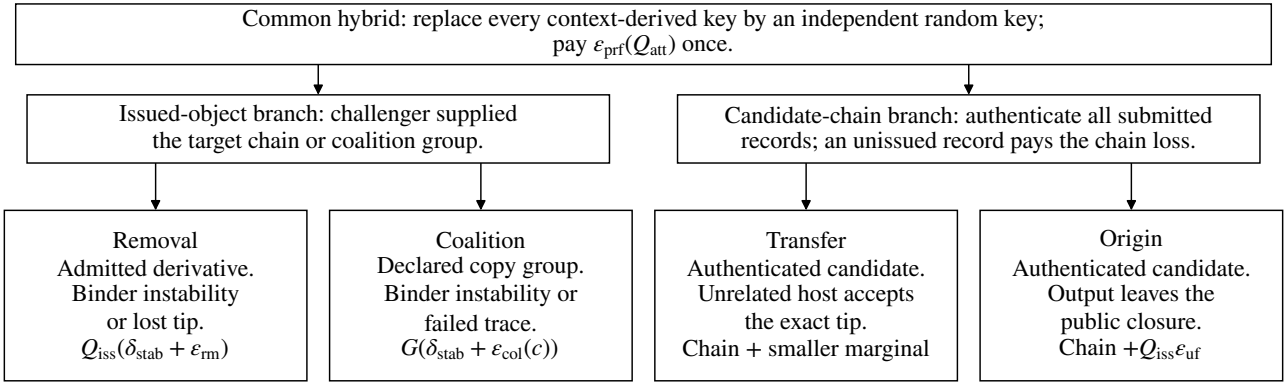
Common hybrid: replace every context-derived key by an independent random key;
pay $\varepsilon_{\mathrm{prf}}(Q_{\mathrm{att}})$ once.
Issued-object branch: challenger supplied the target chain or coalition group.
Candidate-chain branch: authenticate all submitted records; an unissued record pays the chain loss.
Removal
Admitted derivative.
Binder instability or lost tip.
$Q_{\mathrm{iss}}(\delta_{\mathrm{stab}} + \varepsilon_{\mathrm{rm}})$
Coalition
Declared copy group.
Binder instability or failed trace.
$G(\delta_{\mathrm{stab}} + \varepsilon_{\mathrm{col}}(c))$
Transfer
Authenticated candidate.
Unrelated host accepts the exact tip.
Chain + smaller marginal
Origin
Authenticated candidate.
Output leaves the public closure.
Chain $+Q_{\mathrm{iss}}\varepsilon_{\mathrm{uf}}$